

## Challenge (Habanero)

**Q1.** The  $x^2 + 8x + 12$  equation represents the population of rabbits in a forest. Convert this to vertex form.

(A)  $y = (x + 4)^2 - 4$

(B)  $y = (x + 4)^2 + 4$

(C)  $y = (x + 3)^2 - 3$

(D)  $y = (x + 6)^2 - 6$

(E)  $y = (x - 4)^2 + 4$

(E)  $C = (x - 1)^2 + 1$

**Q2.** Complete the square for  $x^2 + 5x$  to find the vertex form.

(A)  $y = (x +$

$\frac{5}{2})^2 -$

$\frac{25}{4}$

(A)  $y = (x -$

$\frac{5}{2})^2 +$

$\frac{25}{4}$

(A)  $y = (x + 2)^2 - 1$

(B)  $y = (x -$

$\frac{5}{2})^2 -$

$\frac{25}{4}$

(B)  $y = (x +$

$\frac{5}{2})^2 +$

$\frac{25}{4}$

**Q3.** A hiker's distance  $d$  from the campsite after  $t$  hours is  $d = t^2 - 6t + 5$ . Rewrite this equation in vertex form.

(A)  $d = (t - 3)^2 - 4$

(B)  $d = (t + 3)^2 + 4$

(C)  $d = (t - 2)^2 - 1$

(D)  $d = (t + 2)^2 - 1$

(E)  $d = (t - 4)^2 + 5$

**Q4.** The cost  $C$  of transporting  $x$  kilograms of cargo is  $C = x^2 + 4x + 4$ . Rewrite this equation in vertex form.

(A)  $C = (x + 2)^2$

(B)  $C = (x - 2)^2$

(C)  $C = (x + 1)^2 + 3$

(D)  $C = (x + 3)^2 - 2$

**Q5.** The area of a plot of land is given by  $A = x^2 + 2x - 6$ . Convert this to vertex form.

(A)  $A = (x + 1)^2 - 7$

(B)  $A = (x - 1)^2 + 7$

(C)  $A = (x + 2)^2 - 2$

(D)  $A = (x - 2)^2 + 2$

(E)  $A = (x + 3)^2 - 3$

**Q6.** Complete the square for  $x^2 - 3x - 2$  to find the vertex form.

(A)  $y = (x -$

$\frac{3}{2})^2 -$

$\frac{17}{4}$

(A)  $y = (x +$

$\frac{3}{2})^2 +$

$\frac{17}{4}$

(A)  $y = (x - 2)^2 + 1$

(B)  $y = (x + 2)^2 - 1$

(C)  $y = (x -$

$\frac{3}{2})^2 +$

$\frac{17}{4}$

**Q7.** A fuel consumption model is given by  $F = x^2 - 4x + 3$ . Rewrite this equation in vertex form.

(A)  $F = (x - 2)^2 - 1$

(B)  $F = (x + 2)^2 + 1$

(C)  $F = (x - 1)^2 + 2$

(D)  $F = (x + 1)^2 - 2$

(E)  $F = (x - 3)^2 + 3$

**Q8.** Complete the square for  $x^2 + 2x - 1$  to find the vertex form.

(A)  $y = (x + 1)^2 - 2$

(B)  $y = (x - 1)^2 + 2$

(C)  $y = (x + 2)^2 - 1$

(D)  $y = (x - 2)^2 + 1$

(E)  $y = (x + 1)^2 + 2$

## Open Ended Questions (FRQ)

**Q9.** A park ranger wants to create a rectangular trail with an area of  $A = x^2 + 5x + 6$ . Convert this equation into vertex form and interpret the results in the context of the trail's dimensions.

**Q10.** The population  $P$  of a certain species of bird is modeled by the equation  $P = t^2 - 8t + 12$ , where  $t$  is the number of years. Rewrite this equation in vertex form and describe the significance of the vertex in this context.

## Chef's Tips

- Emphasize the importance of completing the square for vertex form conversion.
- Use real-world examples to illustrate the relevance of vertex form in modeling.
- Encourage students to interpret the results of vertex form conversion in context.